

Other companies find favour within a grid network. When Ford Motors was in the market for an affordable solution to simulate car chassis behaviour, Sun Microsystems pitched in, with Gridware in tow. Gridware, a German company, now acquired by Sun, made software that tied desktop computers into a mini-grid, able to perform like a supercomputer. The solution saved Ford \$100 million and success thus spawned, encouraged the motor company to use the same solution to design other components.

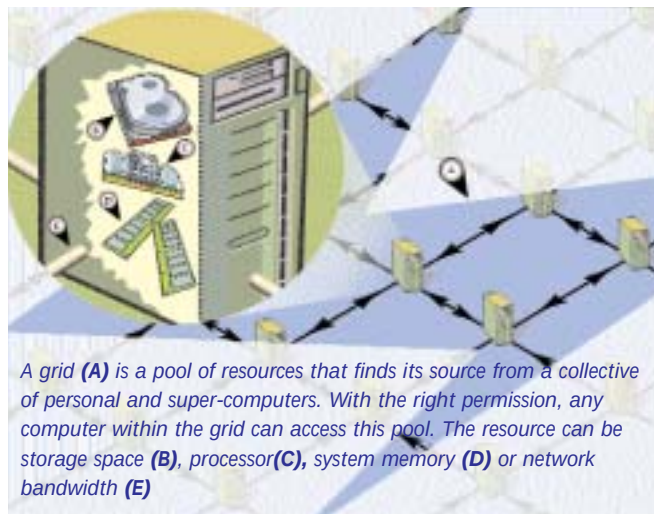
Coming soon...

BMW, Boeing and Motorola have also deployed mini-grids as an affordable solution to enable virtual organisations that span nations and yet provide transparent, immediate access to its resources.

To consumers and home users, the trickle down of this technology is minimal. While distributed services such as SETI@Home are here for desktop PCs, they are hardly meant to grant immediate benefits to an end-user. Perhaps Microsoft with its much ballyhooed .NET initiative or IBM spearheading the rival and open-source OGSA (Open Grid Services Architecture, a set of standards and technologies to grid-enable computers using Internet standards already in place) will trigger a wave of grid-ready applications. Already McAfee.com Corp has launched a new

Butterfly Grid is powered by Linux-based IBM machines hosted by Big Blue and connected via efficient fibre-optics. The grid design potentially allows for one million simultaneous players to game together, as opposed to the several thousand that would conventionally fit into one server. The trick is to grid-connect multiple servers then offer a one-point gateway to the collective for the desktop PCs running a game. The gateway then seamlessly toggles between constituent servers as and when required—to load balance or provide automatic fail-over capability.

The computational horsepower that games demand has fuelled rumours that Sony is looking to a distributed network to leverage the power of the PlayStation 3 (PS3). Grid-enabled (likely with the use of proprietary software), a collection of PS3s will allow for transparent gaming all over the world. The computing resource this will unlock certainly promises titles remarkably realistic in look and feel.



slowly merging into a unified point of view: open-source, community-driven and corporation-backed initiatives such as the Globus project. A further need to tackle issues of security and rights management is urgently felt. Scaglia of Hewlett-Packard has been quoted as saying that the current level of security offered by a grid environment is not acceptable. Perhaps the biggest hurdle is the economic practicality. Grid needs to be largely accepted by organisations with deep pockets. The NSFs, the IBMs and the Microsofts of the world need to take notice. Although the initiative is



It's kind of like the Borg—all the resources become part of the collective, but in a good way

—Tom Hawk on grid computing,
General Manager of grid computing for IBM



initiative called 'Grid Computing Services' that will use distributed computing techniques to provide real-time, dynamic security against viruses or trojans. McAfee's hub will consist of desktop PCs that will process data coming in from users and generate alerts, updates, fixes and information about security threats—in the form of pop-up windows that will either be routed through proprietary software or through Microsoft's .NET Alerts.

Butterfly.net and IBM have teamed up to offer a middleware solution (a middleware lets you deploy a solution with minimal coding effort) for the online video game industry. MMOGs (Massively Multiplayer Online Games) are a lucrative market, fraught with pitfalls involving maintenance of networks and hundreds of thousands of lines of game code. The

There are other promises that tantalise the home segment. Grid-powered servers would mean more dependable Internet access and better Quality of Service, as such a system load-balances itself. Individuals will have faster computers, greater storage capacities and better bandwidth at their fingertips, at costs affordable to both the rich and the poor alike. Cheap hardware such as handhelds (like the Simputer, theoretically) could tap into the power of a grid to offer experiences exponentially greater than otherwise possible.

...right after this commercial break

But there are hurdles. The technology is new and as such there is some uncertainty on the best route to take. A fractured vision of 'how to get there' is

heading towards that goal, the flow of green needs to be much more generous than it currently is. That and co-operation rather than competition, to define overlapping open-source standards and then adopting them remains the key.

We are witnesses to an information overload. Every day massive amounts of data flow helter-skelter, going-a-begging to be structured and analysed, to be made sense of. Hidden beneath this stream could perhaps run cure for diseases: there are 10-100 million species on this planet, 1.7 million are known, and only 50,000 have been studied in any detail. A wealth of knowledge lies at our feet and the limited brain can only look to the tools it has built for a feasible solution. Grid computing then seems the logical step forward. ■

AHMED SHAIKH